Week 10 (CS 418 @ UIC)

Week 10 Slide Deck

Clustering, PCA; Model Evaluation

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Topic Title Placeholder

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Ashland
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Clustering

What Is Clustering?

Unsupervised learning — discover what is already there.

- No labels / ground truth
- Group similar observations together
- Use similarity / distance to make groups
- Goals:
  - high similarity within clusters (cohesion)
  - low similarity between cluster (separation)

Types of Clustering

- **Complete vs. partial** — every observation belongs to a cluster, or some may not
- **Non-overlapping vs. overlapping** — each observation belongs to one cluster, or potentially many

A **clustering** is a collection of clusters; different algorithms make different assumptions about these properties.

K-Means Clustering

Algorithm sketch:

1. Choose k (number of clusters)
2. Initialize k centroids randomly
3. Assign each point to the nearest centroid
4. Recompute centroids as the mean of assigned points
5. Repeat 3–4 until assignments stabilize

Sensitive to initialization; run multiple times and keep the best result.

Choosing K

- **Elbow method**
- **Silhouette score**
- **Domain knowledge**
- ?

K-Means Limitations

- Assumes spherical clusters, roughly equal size
- Sensitive to outliers
    - centroid gets pulled
- Must choose k in advance
- Poor for non-convex shapes
    - DBSCAN, hierarchical clustering as alternatives

K-Means Initialization

K-means is **sensitive to initialization**

Furthest-First Heuristic (FFH):

1. Pick first centroid randomly from data points
2. For each remaining centroid: pick the point that maximizes minimum distance to already-chosen centroids

K-means++: randomized version of FFH — pick next centroid with probability proportional to its minimum distance from existing centroids.

Cluster Evaluation

Cohesion and Separation

- **Cohesion** — how similar observations within a cluster are to each other
- **Separation** — how distinct a cluster is from other clusters

$$\text{cohesion}(C_i) = \frac{1}{|C_i|(|C_i| - 1)/2} \sum_{x,y \in C_i} \text{proximity}(x, y)$$

$$\text{separation}(C_i, C_j) = \frac{1}{|C_i||C_j|} \sum_{x \in C_i,\, y \in C_j} \text{proximity}(x, y)$$

We want high cohesion and high separation.

Silhouette Coefficient

Combines cohesion and separation into a single score per observation:

$$s_i = \frac{b - a}{\max(a, b)}$$

- a = average distance from point i to all other points **in the same cluster**
- b = average distance from point i to all points **in the nearest other cluster**

Range: $[-1, 1]$. Negative → misclassified; 0 → overlapping; positive and close to 1 → well-clustered.

Hierarchical Clustering

Hierarchical Clustering

Produces a **dendrogram** — a tree of nested clusters.

Two approaches:

- **Agglomerative (bottom-up):** start with each observation as its own cluster; merge the closest pair at each step until one cluster remains
- **Divisive (top-down):** start with one cluster; split at each step until each observation is its own cluster

Agglomerative is far more common in practice.

Agglomerative Linkage Methods

How to define "closeness" between clusters:

Method	Distance between C_i and C_j
Single link (MIN)	min distance between any two points
Complete link (MAX)	max distance between any two points
Average link	mean distance across all pairs

TODO look into Ward's method

Average link is most widely used (more robust to outliers than single/complete).

Principal Component Analysis (PCA)

What Is PCA?

Dimensionality reduction — represent high-dimensional data in fewer dimensions, preserve as much variance as possible.

- Find directions (“principal components”) of maximum variance
- Each PC is a linear combination of original features
- PCs are orthogonal / uncorrelated

Why Use PCA?

- Visualization (project to 2D/3D for plotting)
- Noise reduction (drop low-variance components)
- Speed up downstream models (fewer features)
- Handle multicollinearity

Tradeoff: interpretability — PCs are abstract combinations of original features.

Scree Plot

Plot the **explained variance ratio** for each component. Helps decide how many PCs to keep.

[TODO: add scree plot example]

PCA

Pete Crow-Armstrong, Chicago Cubs CF

#4 · CF · Chicago Cubs

- 2025 NL All-Star, Gold Glove winner
- Hit for the cycle on June 15, 2026
  - I was there!
  - And I thought… PCA with PCA?

Photo: Minda Haas Kuhlmann, CC BY 2.0, via Wikimedia Commons

Sources

1. GitHub source: https://github.com/jackbandy/data-science-fun/blob/main/docs/slides/week10.md.
2. Last modified and compiled June 22, 20:11h Central (Chicago) time.
3. Slides developed using materials from Elena Zheleva and Gonzalo Bello Lander, the Berkeley DS 100 team, Marine Carpuat, and Brian Ziebart.
4. Slide deck built with Quarto revealjs.
5. Title font is Big Shoulders; Body font is Libre Franklin.